

University of Cape Town
Department of Mathematics and Applied Mathematics

2IA Deferred Exam
January, 2023
Time: 2 hours
Total marks: 85, Full marks: 80

- Read the instructions carefully.
- You have 2hrs to complete the exam.
- Make sure that no pages or questions are missing.
- Show all your work or reasoning. Answers without work or reasoning shown will receive no mark.

Surname: _____

First name: _____

Student number: _____

Signature: _____

Do not write below this line:

Question:	1	2	3	4	5	Total
Marks:	15	25	15	15	15	85
Score:						

1. Let G be an arbitrary group and $a, b \in G$.

(a) 10 marks Use induction to prove that $(aba^{-1})^n = ab^n a^{-1}$ for all $n \geq 1$.

(b) 5 marks Use (a) to show that $(aba^{-1})^k = ab^k a^{-1}$ for all negative integers k .

Solution: (a) For $n \geq 1$, let p_n be the statement $(aba^{-1})^n = ab^n a^{-1}$. It is clear that p_1 is true, since $(aba^{-1})^1 = aba^{-1}$ and $ab^1 a^{-1} = aba^{-1}$. Suppose now that p_k is true, that is, $(aba^{-1})^k = ab^k a^{-1}$ and let us show that p_{k+1} is also true. In fact:

$$\begin{aligned} (aba^{-1})^{k+1} &= (aba^{-1})^k (aba^{-1}) \stackrel{\text{(I.H.)}}{=} (ab^k a^{-1})(aba^{-1}) = ab^k (a^{-1}a) ba^{-1} \\ &= ab^k ba^{-1} = ab^{k+1} a^{-1} \end{aligned}$$

By the Principle of Mathematical Induction, we obtain that p_n is true for all $n \geq 1$.

(b) Let k be a negative integer and consider the positive integer $n = -k$. Applying (a) to n we obtain that

$$\begin{aligned} (aba^{-1})^n = ab^n a^{-1} &\Rightarrow (aba^{-1})^{-k} = ab^{-k} a^{-1} \Rightarrow ((aba^{-1})^{-k})^{-1} = (ab^{-k} a^{-1})^{-1} \\ &\Rightarrow (aba^{-1})^k = (a^{-1})^{-1} (b^{-k})^{-1} a^{-1} = ab^k a^{-1}, \end{aligned}$$

as desired.

2. In each case determine whether the statement is true or false. **You will get zero marks if you do not provide an argument supporting your answer.**

- (a) 5 marks Suppose that m and n are two positive integers. If $d > 1$ satisfies that $d = am + bn$ for some integers a and b , then $d = \gcd(m, n)$.
- (b) 5 marks The set of all permutations S_4 is an abelian group.
- (c) 5 marks Every finite group is cyclic.
- (d) 5 marks Every cyclic group is finite.
- (e) 5 marks If H and K are two subgroups of a group G , then their union $H \cup K$ is also subgroup of G .

Solution: (a) **False:** The converse of Bézout's Lemma does not hold. Take for example, $m = 4$, $n = 8$. Then $\gcd(4, 8) = 4$ and 12 satisfies that $12 = 1 \cdot 4 + 1 \cdot 8$.

(b) **False:** The transpositions $\alpha = (1 \ 2)$ and $\beta = (1 \ 3)$ are in S_4 and they do not commute. In fact:

$$\begin{aligned}\alpha\beta &= (1 \ 2)(1 \ 3) = (1 \ 3 \ 2), \\ \beta\alpha &= (1 \ 3)(1 \ 2) = (1 \ 2 \ 3).\end{aligned}$$

(c) **False:** S_3 is finite and it is not cyclic. Notice that S_3 can not be cyclic because cyclic groups are abelian and S_3 is not abelian.

(d) **False:** $\mathbb{Z}$ is an infinite (additive) cyclic group.

(e) **False:** Take for example $G = \mathbb{Z}$ and $H = 2\mathbb{Z}$ and $K = 3\mathbb{Z}$. Then $H \cup K$ is not a subgroup of $\mathbb{Z}$, since $2, 3 \in H \cup K$ but $2 + 3 = 5 \notin H \cup K$, since it is not a multiple of 2 or 3.

3. Consider the additive cyclic group $\mathbb{Z}_{20}$.

- (a) 10 marks Find all the subgroups of $\mathbb{Z}_{20}$.
 (b) 5 marks Draw the lattice diagram of $\mathbb{Z}_{20}$.

Solution: (a) To determine the subgroups of $\mathbb{Z}_{20}$, we use the Fundamental Theorem of Finite Cyclic Groups, which tells us that $\mathbb{Z}_{20}$ has exactly 6 subgroups of orders the positive divisor of 20, namely, 1, 2, 4, 5, 10, 20. The subgroups of orders 1 and 20 are clearly $\{\bar{0}\}$ and $\mathbb{Z}_{20}$, respectively.

To determine the unique subgroup of the rest of those orders, recall that the order of the subgroup $\langle \bar{d} \rangle = d\mathbb{Z}_{20}$ is the order of the element $\bar{d} = d\bar{1}$, which is given by the formula $o(d\bar{1}) = 20/d$, provided $d|20$.

$$2 = o(d\bar{1}) = \frac{20}{d} \Rightarrow d = 10; \text{ the subgroup of order 2 is } \langle \bar{10} \rangle = 10\mathbb{Z}_{20};$$

$$4 = o(d\bar{1}) = \frac{20}{d} \Rightarrow d = 5; \text{ the subgroup of order 4 is } \langle \bar{5} \rangle = 5\mathbb{Z}_{20};$$

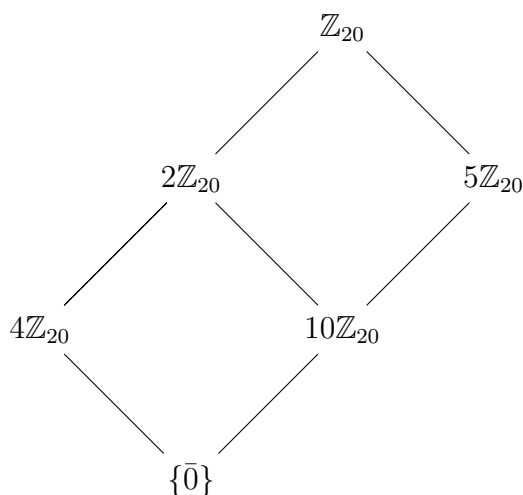
$$5 = o(d\bar{1}) = \frac{20}{d} \Rightarrow d = 4; \text{ the subgroup of order 5 is } \langle \bar{4} \rangle = 4\mathbb{Z}_{20};$$

$$10 = o(d\bar{1}) = \frac{20}{d} \Rightarrow d = 2; \text{ the subgroup of order 10 is } \langle \bar{2} \rangle = 2\mathbb{Z}_{20};$$

(b) To draw the lattice diagram we use that

$$\langle \bar{k} \rangle \subseteq \langle \bar{m} \rangle \Leftrightarrow m|k$$

Thus the lattice diagram of $\mathbb{Z}_{20}$ can be pictured as follows:



4. (a) 5 marks State **Lagrange's Theorem**.
- (b) 4 marks Complete the missing parts (marked with a $\star$) in the statement of the **Fundamental Theorem of Finite Cyclic Groups**.

FUNDAMENTAL THEOREM OF FINITE CYCLIC GROUPS. Let $G = \langle g \rangle$ be a cyclic group of order n .

- (1) If H is a subgroup of G , then $H = \langle g^d \rangle$ for some $(\star)$.
 (2) If H is any subgroup of G with order $|H| = k$, then $(\star)$.
 (3) If $k|n$, then $\langle g^{n/k} \rangle$ is $(\star)$ of G of order $(\star)$.

- (c) Consider the following incomplete proof of (3) in the **Fundamental Theorem of Finite Cyclic Groups**:

Suppose that K is any subgroup of G of order k . Then K is of the form $K = (A)$ by (1). Then using Lemma 3.31 (from class) we obtain that

$$(B) = o(g^m) = \frac{n}{m},$$

and so (C), which shows that $K = \langle g^{\frac{n}{k}} \rangle$, finishing the proof of (3).

With reference to this partial proof:

- i. 2 marks Complete the sentence at (A).
 ii. 2 marks Complete the sentence at (B).
 iii. 2 marks Complete the sentence at (C).

Solution: (a) **Lagrange's Theorem.** Let H be a subgroup of a finite group G . Then the order $|G|$ is a multiple of the order of H .

(b) FUNDAMENTAL THEOREM OF FINITE CYCLIC GROUPS. Let $G = \langle g \rangle$ be a cyclic group of order n .

- (1) If H is a subgroup of G , then $H = \langle g^d \rangle$ for some $d|n$.
 (2) If H is any subgroup of G with order $|H| = k$, then $k|n$.
 (3) If $k|n$, then $\langle g^{n/k} \rangle$ is the **unique subgroup** of G of order k .

(c)

- i. $K = \langle g^m \rangle$, for $m|n$.
 ii. $k = |K| = o(g^m) = \frac{n}{m}$
 iii. $m = \frac{n}{k}$.

5. Let G be a group and H a normal subgroup of G . Suppose that K is a normal subgroup of H . For $a \in G$, prove that

- (a) 8 marks $aKa^{-1} = \{axa^{-1} \mid x \in K\}$ is a subgroup of H .
 (b) 7 marks aKa^{-1} is a normal subgroup of H .

Hint! In class, we proved that a subgroup A of a group G is normal if and only if $gAg^{-1} \subseteq A$ for all $g \in G$ (*Normality Test*). One can show that

a subgroup A is normal if and only if $gAg^{-1} = A$ for all $g \in G$.

Please, use this new version of the Normality to prove (b).

Solution: (a) Let us begin by noticing that $aKa^{-1} \subseteq H$ because $K \subseteq H$ and H is normal in G , by hypothesis.

To check that aKa^{-1} is a subgroup of H we will use the Subgroup Test:

- $1 \in aKa^{-1}$ since $1 = a1a^{-1}$ (notice that $1 \in K$);
- if $ax_1a^{-1}, ax_2a^{-1} \in aKa^{-1}$ (for some $x_1, x_2 \in K$) then

$$(ax_1a^{-1})(ax_2a^{-1}) = ax_1(a^{-1}a)x_2a^{-1} = a(x_1x_2)a^{-1} \in aKa^{-1},$$

since $x_1x_2 \in K$;

- if $axa^{-1} \in aKa^{-1}$ (for some $x \in K$) then

$$(axa^{-1})^{-1} = (a^{-1})^{-1}x^{-1}a^{-1} = ax^{-1}a^{-1} \in aKa^{-1},$$

since $x^{-1} \in K$.

(b) From (a) we know that aKa^{-1} is a subgroup of H . To show that aKa^{-1} is normal in H , we use the Normality Test (version provided in the hint). For $h \in H$ we need to check that $h(aKa^{-1})h^{-1} = aKa^{-1}$. Let us begin by noticing that $yKy^{-1} = K$, for all $y \in H$ since K is normal in H (using the Normality Test provided in the hint here). In particular, for $y = h^{-1}$, we get

$$K = h^{-1}K(h^{-1})^{-1} = h^{-1}Kh.$$

From here we obtain that

$$\begin{aligned} h(aKa^{-1})h^{-1} &= (ha)K(a^{-1}h^{-1}) = (ha)(h^{-1}Kh)(a^{-1}h^{-1}) \\ &= (hah^{-1})K(ha^{-1}h^{-1}) = (hah^{-1})K(hah^{-1})^{-1} = K, \end{aligned}$$

since K is normal in G , by hypothesis.